Simplificar

A1

$$\frac{x}{(x+y)^2} - \frac{y}{(x-y)^2} - \frac{x+y}{x^2-y^2}$$

Solución:

$$\frac{x(x-y)^2 - y(x+y)^2 - (x+y)(x-y)(x+y)}{(x+y)^2(x-y)^2}$$

$$\frac{x(x^2-2xy+y^2) - y(x^2+2xy+y^2) - (x^2-y^2)(x+y)}{(x+y)^2(x-y)^2}$$

$$\frac{\cancel{x^3} - 2x^2y \cancel{+ xy^2} - x^2y \cancel{- 2xy^2} \cancel{- y^3} \cancel{- x^3} - x^2y \cancel{+ xy^2} \cancel{+ y^3}}{(x+y)^2(x-y)^2}$$

$$\frac{4x^2y}{(x+y)^2(x-y)^2}$$

$$-\frac{4x^2y}{[(x+y)(x-y)]^2}$$

$$\boxed{-\frac{4x^2y}{(x^2-y^2)^2}}$$ Respuesta

c

Solución: $\dfrac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}} = 1$

$\boxed{A2}$

$$\cancel{\sqrt{1+x}} + \sqrt{1-x} = \cancel{\sqrt{1+x}} - \sqrt{1-x}$$

$$2\sqrt{1-x} = 0 \; ; \quad \left(\sqrt{1-x}\right)^2 = (0)^2$$

$$1-x = 0 \; ; \quad x = 1$$

\# de soluciones $\boxed{1}$ B

Solución:

A3

$$\begin{cases} \dfrac{2x}{m+n} + \dfrac{2y}{m-n} = \dfrac{2}{m+n} \\ \dfrac{2x}{m+n} - \dfrac{2y}{m-n} = \dfrac{2}{m-n} \end{cases}$$

$$\frac{4x}{m+n} = \frac{2}{m+n} + \frac{2}{m-n}$$

$$\frac{4x}{m+n} = \frac{2m-2n+2m+2n}{(m+n)(m-n)} \; ; \; \frac{4x}{m+n} = \frac{4m}{(m+n)(m-n)}$$

$$x = \frac{m}{m-n}$$

$$\begin{cases} \dfrac{2x}{m+n} + \dfrac{2y}{m-n} = \dfrac{2}{m+n} \\ -\dfrac{2x}{m+n} + \dfrac{2y}{m-n} = \dfrac{-2}{m-n} \end{cases}$$

$$\frac{4y}{m-n} = \frac{2}{m+n} - \frac{2}{m-n} \; ; \; \frac{4y}{m-n} = \frac{2m-2n-2m-2n}{(m+n)(m-n)}$$

$$\frac{4y}{m-n} = \frac{-4n}{(m+n)(m-n)} \; ; \quad y = -\frac{n}{m+n}$$

$$x+y = \frac{m}{m-n} - \frac{n}{m+n} = \frac{m^2+mn-mn+n^2}{(m+n)(m-n)}$$

$$\boxed{x+y = \frac{m^2+n^2}{m^2-n^2}}$$

c

Solución: x número de salteñas

$\boxed{A4}$

1ro: $x-70 > \frac{x}{2}$

2do: $(x-70)+8-38 < 42$

Resolviendo: $2x-140 > x$; $x-140 > 0$

$x > 140$

$x - 100 < 42$; $x < 142$

Luego: $140 < x < 142$

Se tiene que $\boxed{x = 141}$ B

Solución

$\boxed{A5}$

$$\frac{8x^2-12}{x^3+2x^2-8x} = \frac{8x^2-12}{x(x^2+2x-8)}$$

$$= \frac{8x^2-12}{x(x+4)(x-2)}$$

$$= \frac{A}{x} + \frac{B}{x+4} + \frac{C}{x-2}$$

lo cual no tiene la forma $\frac{A}{x} + \frac{B}{x-4} + \frac{C}{x+2}$

por lo tanto la respuesta es:

E